

Task 3 – Systemic Outlook and Communication

Group Plastics in agriculture - 4

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1. Global and Local Significance of the Practices

Mulch film choices in organic strawberry production are not merely a local farming decision – they carry global sustainability implications. Whether a system uses LDPE, biodegradable plastics, or no mulch affects greenhouse gas emissions, fossil resource use, and plastic pollution worldwide. A key finding from our assessment is that the biodegradable option (S4), often assumed to be environmentally preferable, actually generates the highest climate impact (0.242 kg CO_2-eq; Task 2, Table 1) because of energy-intensive feedstock processing and field-stage degradation. This result is locally relevant for Nordic growers under pressure to adopt bio-based films and globally significant for policies that promote biodegradable materials without full life-cycle scrutiny (Razza & Cerutti, 2017).

2. Key Trade-Offs Across Sustainability Criteria

Three fundamental trade-offs emerged from the quantitative comparison.

2.1 Environmental versus economic performance

S1 (no mulch) leads all environmental and health indicators but generates zero profit and saves no labour (Task 2, Fig. 1). Without economic returns, S1 cannot support farm livelihoods. S4 offers the highest profitability (€12,920/ha) and the sole labour saving (50 h/ha), yet carries the heaviest environmental and toxicity burdens (Task 2, Fig. 2) – a clear "People–Planet–Profit" tension.

2.2 Trade-offs within the environmental dimension

Climate change and fossil resource depletion do not fully align: S4's CO_2-eq is markedly higher than S2/S3, while its fossil resource use (2.87 MJ) is only moderately above LDPE values (~2.06 MJ; Task 2, Table 1). Indicators excluded from the core set could further shift rankings – biodegradable films often show elevated eutrophication during degradation, which would further weaken S4's environmental profile.

2.3 Recycling's impact is systemic, not chemical

S2 and S3 are nearly identical in all emission-based indicators (Task 2, Fig. 1). Recycling does not meaningfully lower production-stage impacts, but it addresses end-of-life plastic accumulation – a benefit our indicator set does not directly capture. This mismatch highlights

a limitation of the assessment.

3. Most Important Challenges and Decision-Making Methods

No single scenario excels across all dimensions: farmers favour S4 for income, regulators S1 for climate, circular economy proponents S3. Multi-criteria decision analysis (MCDA) can resolve such conflicts by letting stakeholders assign weights (DTU Sustain, 2026), but weightings are inherently subjective, making transparency about value choices essential. Structured workshops can help define acceptable trade-off zones.

4. Major Assumptions and Uncertainties

Several assumptions and data gaps significantly affect our conclusions.

4.1 Indicator selection

We selected five of 23 available indicators. Choosing water consumption or marine eutrophication instead of fossil resource use would likely worsen S4's position; emphasising land use efficiency could weaken S1 due to lower per-hectare yield. The final ranking is therefore sensitive to indicator choice (Task 2, Section B1).

4.2 Absence of microplastics indicators

The database lacks a microplastics indicator. If included, S2 would be penalised most by film fragmentation and soil accumulation, while S3 would benefit from film diversion through recycling. S4's incomplete biodegradation could generate secondary microplastics with unknown ecological effects (Abeynayaka, 2026). Microplastic inclusion would thus widen the gap between S2 and S3 and raise new concerns about S4's long-term safety.

4.3 System boundaries and soil dynamics

Our cradle-to-grave scope excludes capital goods and long-term soil health indicators such as soil organic carbon and soil biodiversity. If soil carbon sequestration under biodegradable mulch were fully accounted, S4 might improve marginally, though current evidence suggests only small benefits. Soil biodiversity, likely highest under S1 and lowest under S2 (where LDPE fragments persist), could shift the ranking if quantified.

4.4 Yield and degradation assumptions

The underlying data assume identical yields for all mulched scenarios. If biodegradable film degrades prematurely in warmer climates, S4's economic advantage would shrink, potentially making S3 the more robust choice.

4.5 Expectation vs. data

We initially expected biodegradable film to have the lowest carbon footprint. The data contradicted this expectation (Task 2, Section A1), a reminder that life-cycle evidence is essential for material comparisons.

5. Other Important Indicators Not Included

Beyond microplastics, the assessment lacks quantitative treatment of soil biodiversity, soil organic carbon change, and social acceptance. Soil biodiversity is likely to be highest under S1 (no plastic input) and lowest under S2 (accumulated LDPE fragments). Social acceptance, explored qualitatively in Task 1, is equally decisive: farmers must trust that recycling infrastructure will reliably accept used films and that biodegradable mulches will perform consistently. Without this trust, even the highest-scoring scenario on paper may fail in the field.

6. Circular Economy Practices: Impact and Barriers

Among circular strategies, mechanical recycling of LDPE film (S3) offers the greatest systemic benefit by reducing plastic leakage into soils and lowering virgin polymer demand, even though upstream impact reductions remain modest (Task 2, Fig. 1, S2 vs. S3). Composting of biodegradable film (S4) is limited by insufficient field degradation conditions, risking incomplete breakdown and compost contamination (Razza & Cerutti, 2017). Reuse is constrained by film tearing and soil adhesion.

The main barriers to S3-type recycling are collection logistics, soil contamination, and a lack of nearby facilities that accept agricultural film. Because farmers gain no labour-saving benefit from careful removal (Task 2, Fig. 1, labour panel), financial incentives or extended producer responsibility (EPR) schemes are needed. Subsidised collection points, recyclate quality standards, and farmer training could help, but these require coordinated action across the value chain. Without such measures, S3 remains a theoretical optimum rather than a field reality.

References

1. Abeynayaka, A. (2026). *Agri-Plastics Case Study – Task 3 Lecture Slides*. DTU Sustain, Technical University of Denmark.

2. Razza, G. & Cerutti, A. (2017). Life cycle assessment of biodegradable and non-biodegradable mulch films. (Cited in course lectures, Task 2.)

3. European Commission. (2020). *Sustainable Product Policy Initiative*. Brussels: European Commission.

APPENDIX B: Task 1: Problem Formulation – Sustainability of Agricultural Mulching Practices for Strawberry Cultivation

(A)Crop Selection

Selected crop: Strawberry

(B). Selected Agricultural Practices (4 Scenarios)

(B1)From the 14 available organic open-field scenarios, four representative systems were selected to explore two main questions:

- How does irrigation influence sustainability?
- How does mulch material and end-of-life (EoL) management influence environmental performance?

(B2)Selected Scenarios:

- Organic cultivation without mulching, without irrigation**(S1)**
- Organic cultivation with LDPE mulchfilm, without EoL management, with irrigation**(S2)**
- Organic cultivation with LDPE mulchfilm, with recycling, with irrigation**(S3)**
- Organic cultivation with biodegradable mulchfilm, with irrigation**(S4)**

(B3)These scenarios allow comparison between:

- No mulch vs plastic mulch
- Linear vs circular plastic systems
- Fossil-based vs biodegradable materials

(C) Description of Selected Practices

	S1	S2	S3	S4
Material	No mulching material	Fossil-based LDPE film (20–40 µm thickness)	Fossil-based LDPE film (20–40 µm thickness)	Bio-based and biodegradable mulch film
Production	Not applicable	Polymerization of ethylene derived from oil or natural gas, followed by film extrusion. Energy-intensive and fossil-resource dependent	Polymerization of ethylene derived from oil or natural gas, followed by film extrusion. Energy-intensive and fossil-resource dependent	Biomass cultivation (e.g., starch crops), polymer blending, and extrusion. Production may involve renewable carbon but still requires energy inputs.
Purpose	Baseline system; relies on natural soil conditions.	● Weed suppression ● Soil warming ● Moisture retention ● Increased and earlier yield	● Weed suppression ● Soil warming ● Moisture retention ● Increased and earlier yield	● Weed control ● Soil moisture conservation ● Elimination of removal after use
End-of-life	No mulch waste generated.	No EoL management	Collected and mechanically recycled	Incorporated into soil and designed to biodegrade.

(C1) Effects on Agricultural Inputs and Outcomes

Compared to no-mulch organic cultivation:

	Irrigation	Manual Weeding	Fruit Quality	Yield vs S1	Notes
S1: No mulch, non-irrigated	Baseline (rainfed)	Highest	Lowest	Baseline	Reference for low-input organic
S2/S3: LDPE, irrigated	↓ 30-40%	↓↓ 70-90%	High	↑ 30-50%	Major labor savings
S4: Biodegradable, irrigated	↓ 30-40%	↓↓ 70-90%	High	↑ 30-50%	Similar to LDPE agronomically

(D) Object of Assessment – Functional Unit (FU)

Two functional units were considered:

Option 1: Per kilogram of market-ready strawberries

Reflects agricultural output and productivity differences.

Option 2: Per hectare cultivated land

Common in agricultural statistics but does not account for yield variation.

(D1) Influence of FU Choice

Per hectare comparison may make LDPE systems appear worse due to plastic production burdens.

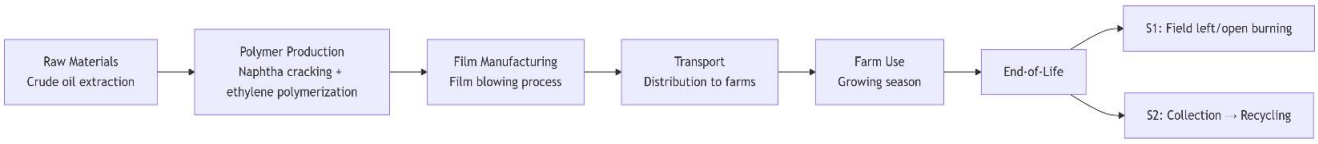
Per kg comparison may favor higher-yield systems because environmental burdens are distributed over more output.

(D2) Selected Functional Unit

1 kg of market-ready organic strawberries is selected as the primary functional unit because agriculture's main function is food production. This unit allows fair comparison between systems with different productivity levels.

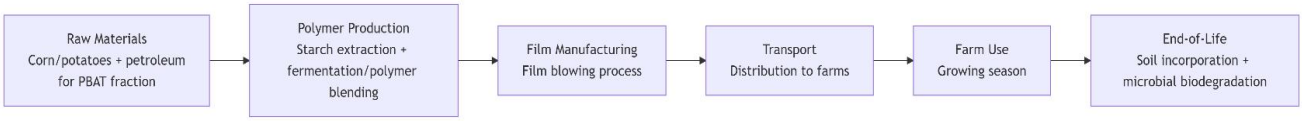
(E) Lifecycle Sketches and Critical Processes Lifecycle Diagrams

LDPE Mulch (S2/S3):



Key impacts: (1)Greenhouse gas emissions (polymer production). (2)Fossil resource depletion. (3)Plastic pollution risk

Biodegradable Film (S4):



Key impact: (1) Land use and agricultural inputs for feedstock. (2)Energy use during processing (3)Soil health implications

No Mulch (S1):



Most Important Processes to Include in Assessment

Based on the analysis, the following processes are critical to include:

- Polymer manufacturing (LDPE & biodegradable)
- Use phase (weed suppression, irrigation reduction)
- End-of-life (especially S2: no management)
- Raw material extraction (crude oil)
- Feedstock cultivation (biodegradable)
- Transport

APPENDIX B: TASK 2 – Sustainability Assessment - Quantification

A). Identify Indicators

A1). Environmental Impacts

Across the life cycle of mulch systems, the most pressing environmental issues are climate change, fossil resource depletion, and chemical toxicity. Eutrophication was excluded because all scenarios follow strict organic fertilisation rules; land use was excluded because yield differences are already captured by the per-kg functional unit.

Indicators: Climate change (kg CO_2-eq), fossil resource use (MJ), and human toxicity non-cancer (CTUh).

Expectation vs. data: We initially expected the biodegradable film (S4) to show the lowest carbon footprint due to its bio-based origin. In reality, S4 carries the highest climate impact (0.242 kg CO_2-eq) because of energy-intensive feedstock processing and emissions during field biodegradation.

Expected ranking: S1 (no mulch) should have the lowest environmental burden; S4 the highest. S2 and S3 (both LDPE) are expected to fall in between, with recycling (S3) bringing only modest improvements.

A2). Health-Related Aspects

The main health concern is non-cancer toxicity from chemical emissions during polymer production and film degradation, as well as occupational risks during manual film removal. Particulate matter and photochemical ozone are also relevant but CTUh offers the most direct

comparison within the available data. Worker acceptance is noted qualitatively.

Indicator: Human toxicity non-cancer (CTUh).

Expected ranking: S1 poses the lowest risk by avoiding plastic-related emissions entirely. S4 is expected to show the highest toxicity, a pattern that the data later confirms. S2 and S3 should be nearly identical in toxicity, as recycling does not alter emission profiles during use.

A3). Socio-Economic Aspects

Farm profitability and labour demand are decisive for whether a practice is adopted at scale. Input security and recycling infrastructure matter, but as enabling conditions rather than direct outcome measures; we therefore focus on direct metrics.

Indicators: Profitability (€/ha) and labour hours saved (h/ha).

Expected ranking: S4 is expected to perform best due to high yields and zero manual removal. S1 is predicted to be the weakest, with no profit and high weeding labour. S2 and S3 should deliver comparable economic returns, with S3 marginally lower due to recycling costs.

B). Quantifying Impacts

B1). Selection of Key Indicators

From the 23 indicators in the database and the discussion above, we keep five that balance environmental, health and socio-economic dimensions while including the mandatory CO_2-eq:

1. Climate change (kg CO_2-eq)
2. Fossil resource use (MJ)
3. Human toxicity non-cancer (CTUh)
4. Profitability (€/ha)
5. Labour hours saved (h/ha)

B2). Quantitative Comparison and Visualisation

Table 1 gives the raw values. Figure 1 then presents all four indicators side-by-side in a normalised multi-panel comparison.

Table 1 – Quantitative comparison per functional unit

Indicator	S1	S2	S3	S4	Best
Climate change (kg CO_2-eq)	0.145	0.178	0.178	0.242	S1
Fossil resource use (MJ)	1.326	2.058	2.061	2.870	S1
Human tox. non-cancer (CTUh)	4.08×10^{-9}	5.25×10^{-9}	5.25×10^{-9}	5.55×10^{-9}	S1
Profitability (€/ha)	0	12,700	12,300	12,920	S4

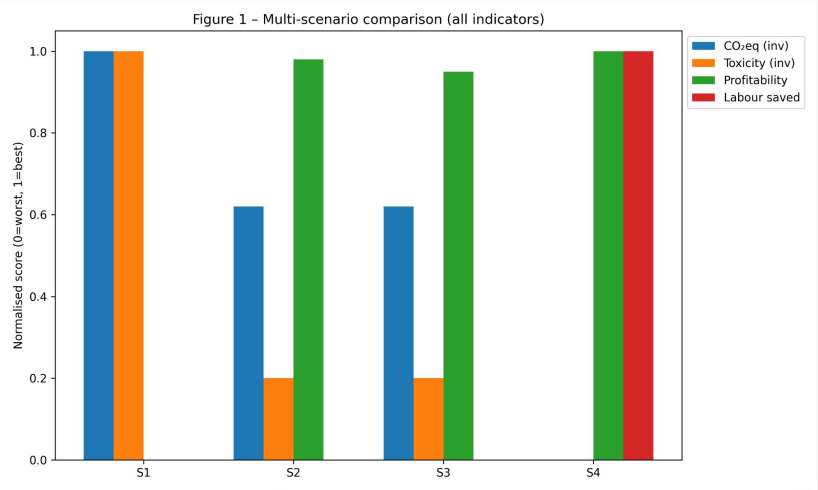


Fig. 1 – Normalised indicator scores (0 = worst, 1 = best). Environmental indicators are inverted so that higher bars always mean better performance. S1 peaks environmentally; S4 peaks socio-economically. S2 and S3 show identical environmental profiles and near-identical economic scores.

A clear environment–economy tension is visible in Figure 1: S1 is the environmental optimum but economically unviable; S4 is the economic optimum but worst for climate and toxicity. S2 and S3 are practically identical in all four panels, which tells us that recycling provides no upstream relief – but it does tackle end-of-life waste, a critical long-term factor that these indicators do not capture.

To compare across dimensions simultaneously, we plotted the same normalised data on a radar chart.

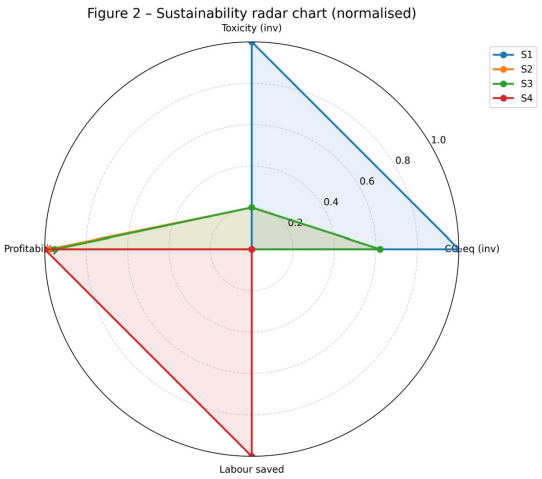


Fig. 2 – Radar chart. S4 peaks socio- economically but lags environmentally. S3 shows the most balanced polygon across all four axes.

We then assigned penalty points based on each scenario's relative weaknesses to obtain an overall rank. The result, from most to least sustainable, is S3 > S1 > S4 > S2 (Figure 3). S3 leads because it matches S2's profitability and environmental footprint, while the integrated recycling loop removes the plastic waste liability that pushes S2 to the bottom. S1's strong environmental scores are dragged down by zero economic returns; S4 is penalised by its high toxicity and carbon footprint despite excellent farmer- friendly attributes.

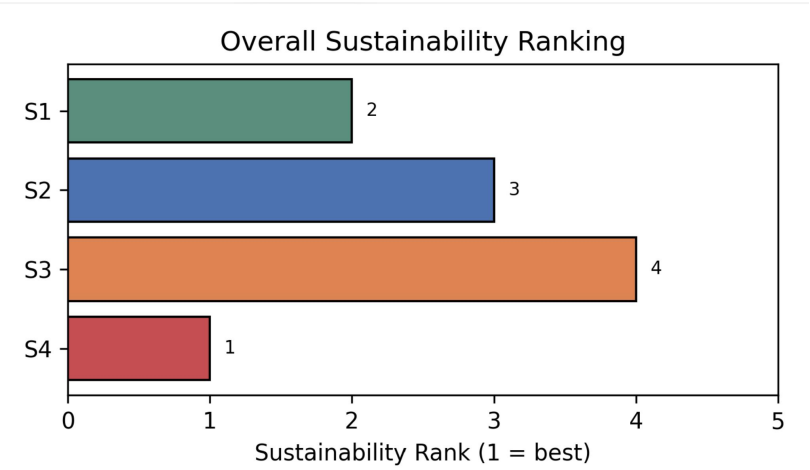


Fig. 3 – Overall sustainability ranking: S3 (1st) > S1 (2nd) > S4 (3rd) > S2 (4th).

B3). Absolute Sustainability Assessment for CO_2- eq

We applied two allocation approaches (DTU Sustain, 2026).

- *Grandfathering:* Using S1's 0.145 kg CO_2-eq as the baseline "right to emit", S3 (0.178 kg) exceeds it by ~22.7 %. All mulching scenarios fail this test.
- *Economic allocation:* Even after distributing emissions according to economic output, all scenarios retain net positive emissions.

None of the four practices are absolutely sustainable from a climate standpoint. However, a strict grandfathering approach would permanently favour the no-mulch system, which is incapable of sustaining farm livelihoods. For food systems, absolute environmental boundaries must be weighed against socio-economic necessities.

B4). Recommended Agricultural Practice

We recommend S3 – LDPE mulch film with irrigation and dedicated recycling. It delivers the best overall sustainability profile: strong profitability (€12,300/ha), environmental impacts comparable to standard LDPE, and a deliberate end-of-life solution that avoids the persistent waste problem of unmanaged plastic. This case demonstrates that when recycling is built into the system, conventional plastic can outperform biodegradable alternatives in holistic sustainability (Razza & Cerutti, 2017).